

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
1	Indian Pines	2000	C	16	HS (Aerial)	1	145	20m	0.006	?	?	?	—
2	Salinas	2000	C	16	HS (Aerial)	1	365	3.7m	0.027	?	?	?	—
3	Botswana	2005	C	14	HS (Aerial)	1	875	30m	0.079	?	?	?	—
4	Kennedy Space Center	2005	C	13	HS (Aerial)	1	550	0.18m	0.057	?	?	?	—
5	Pavia Center	2009	C	9	HS (Aerial)	1	1096	—	0.124	?	?	?	—
6	Pavia University	2009	C	9	HS (Aerial)	1	610	—	0.033	?	?	?	—
7	UC Merced	2010	C	21	RGB	2.1K	256	0.3m	0.317	Yes	?	?	TG
8	WHU-RS19	2012	C	19	RGB	1K	600	0.5m	0.1	?	?	?	—
9	Brazilian Coffee Scenes (BCS)	2015	C	2	RGB	2.9K	64	20m	0.004	?	?	?	—
10	RSSCN7	2015	C	7	RGB	2.8K	400	—	0.086	?	?	?	—
11	SAT-4	2015	C	4	RGB-NIR	500K	28	1~6m	1.15	?	?	?	—
12	SAT-6	2015	C	6	RGB-NIR	405K	28	1m	1.15	?	?	?	—
13	VEDAI	2015	C	9	IRGB	1.2K	1024	0.125m	3.99	Yes	?	?	Paper
14	Brazilian Cerrado-Savanna Scenes (BCSS)	2016	C	4	MS	1.3K	64	5m	0.011	?	?	?	—
15	NWPU-RESISC45	2016	C	45	RGB	31.5K	256	0.2~30m	0.405	Yes	?	?	Paper+TG
16	RSC11	2016	C	11	RGB	1.2K	512	—	—	?	?	?	—
17	SIRI-WHU	2016	C	12	RGB	2.4K	200	2m	0.7	?	?	?	—
18	AID	2017	C	30	RGB	10K	600	3m	2.44	Yes	Yes	?	Paper+REO
19	Austin Zoning	2017	C	5	RGB	3.7K	773x961	—	0.596	?	?	?	—
20	Kag. Planet Forest	2017	C	12	OPT (Satellite)	150K	256	—	33	?	?	?	—
21	MUSIC4GC (Golf Course)	2017	C	2	MS+L	83.4K	16	30m	0.37	?	?	?	—
22	NOAA Sea Lion	2017	OD,C	4	RGB	950	4900	—	96	?	?	?	—
23	RSD46-WHU	2017	C	46	RGB	117K	256	0.5~2m	—	?	?	?	—
24	RSI-CB128	2017	C	45	RGB	36K	128	0.3~3m	0.879	Yes	?	?	Paper
25	RSL-CB256	2017	C	35	RGB	24K	256	0.3~3m	2.24	Yes	?	?	Paper
26	Ships in Satellite Imagery	2017	C	2	RGB	4K	80	3m	0.343	?	?	?	—
27	Solar Power Plants MUSIC4P3	2017	C	2	MS+L	1.3M	16	30m	4.6	?	?	?	—
28	TiSeLaC	2017	C	9	RGB-NIR+MT	23	2866x2633	30m	—	?	?	?	—
29	Cactus Aerial Photos	2018	C	2	RGB	24K	32	—	0.053	Yes	Yes	?	FOMO
30	DeepGlobe-Bldng	2018	S,C	1	OPT (Satellite)	24.6K	650	—	—	Yes	?	?	Paper
31	EuroSAT	2018	C	10	S2 MS	27K	64	10~60m	1.92	Yes	Yes	?	GFM+Paper+TG
32	FMoW	2018	C	63	RGB+MS	523.8K	—	0.3m	3.5K	?	?	?	—
33	Humpback Whale Identification	2018	OD,C	—	OPT (Multiple)	25.5K	1050	—	0.7	?	?	?	—
34	MASATI-v2	2018	C	7	RGB	7.4K	512	0.08~2m	2.3	?	?	?	—
35	MSLCC	2018	C	4	SAR+MS	2	31~957	10m	0.5	?	?	?	—
36	PatternNet	2018	C	38	RGB	30.4K	256	0.062~4.693m	1.3	Yes	?	?	Paper+TG
37	SanFranBay Ships Planet	2018	C	1	OPT (Satellite)	4K	80	—	0.194	?	?	?	—
38	xView	2018	C	60	RGB+MS	1.4K	3000	0.3m	20	Yes	?	?	Paper
39	AIDER	2019	C	4	OPT (Drone)	2.6K	240	—	0.275	?	?	?	—
40	BigEarthNet-MM	2019	C	19	S1/S2	590.3K	20~120	10~60m	121	Yes	Yes	?	FOMO+GFM+Paper
41	DSCR	2019	C	7	RGB	20.7K	150~800	—	—	?	?	?	—
42	HistAerial	2019	C	7	Grayscale	42K	25~100	—	7.6	?	?	?	—
43	ICONES-HSI	2019	C	9	HS (Aerial)	468x224	300	—	11.5	?	?	?	—
44	Kag. Find a Car Park	2019	C	2	RGB	3.3K	1296	—	2.75	?	?	?	—
45	Kaggle Cloud Detection	2019	C	4	RGB	9.2K	1750	—	6	?	?	?	—
46	Kag. Satellite Hurricane Damage	2019	C	—	RGB	16K	128	1m	0.064	?	?	?	—
47	OPTIMAL-31	2019	C	31	RGB	1.9K	256	—	0.024	?	?	?	—
48	SEN12MS	2019	S,C	—	MS+SAR	180.7K	256	10m	510	Yes	Yes	?	FOMO+TG
49	Ship Classification	2019	C	7	OPT (Satellite)	2K	—	—	—	?	?	?	—
50	SloveniaLandCover	2019	C	10	MS	940	—	—	—	?	?	?	—
51	So2Sat LCZ42	2019	C	—	S1+S2	400.7K	32	10m	51.8	Yes	Yes	?	CB+GB2+GFM+Paper
52	USTC SmokeRS	2019	C	6	RGB	6.2K	256	1000m	0.795	?	?	?	—
53	WiDS Datathon 2019	2019	C	2	RGB	11K	256	3m	0.46	?	?	?	—
54	Airbus Wind Turbines Patches	2020	C	1	RGB+MS	358K	128	1.5m	1	?	?	?	—
55	AiRound	2020	C	11	Multi	11.8K	300	—	33	?	?	?	—
56	Bijie Landslide	2020	C	1	RGB	2.8K	200	0.68m	0.513	?	?	?	—
57	BrazilDAM	2020	C	2	RGB	769	384	10~60m	57	Yes	?	?	Paper
58	BreizhCrops	2020	C	9	S2+MT	610K	—	10~60m	—	?	?	?	—
59	CaneSat	2020	C	2	S2+MT	1.6K	10	10m	0.006	?	?	?	—
60	CLRS	2020	C	25	RGB	15K	256	0.26~8.85m	1.74	Yes	?	?	Paper
61	CV4A Kenya	2020	C	7	S2	8.8K	2500	10m	3.5	?	?	?	—
62	CV-BrCT	2020	C	9	RGB	24K	500	—	9.2	?	?	?	—
63	DND-SB	2020	C	7	OPT (Drone)	5.6K	7296	—	49	?	?	?	—
64	FLAME	2020	S,C	2	MS (Drone)	48K	3840	—	43.7	?	?	?	—
65	FUSAR-Ship	2020	OD,C	113	SAR (Satellite)	5K	512	—	—	?	?	?	—
66	IDTreeS	2020	OD,C	33	Multiple (Multiple)	591x369	20	—	0.28	Yes	?	?	TG
67	LandCoverNet Af	2020	C	7	Multiple (Satellite)	2K	256	—	632	?	?	?	—
68	LandCoverNet As	2020	C	7	Multiple (Satellite)	2.8K	256	—	610	?	?	?	—
69	LandCoverNet Au	2020	C	7	Multiple (Satellite)	600	256	—	126	?	?	?	—
70	LandCoverNet Eu	2020	C	7	Multiple (Satellite)	840	256	—	249	?	?	?	—
71	LandCoverNet NA	2020	C	7	Multiple (Satellite)	1.6K	256	—	351	?	?	?	—
72	LandCoverNet SA	2020	C	7	Multiple (Satellite)	1.2K	256	—	215	?	?	?	—
73	MAFAT Challenge	2020	C	—	SAR	—	—	—	—	Yes	?	?	Name
74	MLRSNet	2020	C	46	RGB	109.2K	256	0.1~10m	1.25	?	?	?	—
75	OVERHEAD MNIST	2020	C	9	Grayscale	1K	28	—	0.018	Yes	?	?	Paper
76	Rwanda Crop Type	2020	C	6	OPT (Drone)	6.5K	200	—	56	?	?	?	—
77	S2 Cloud Mask Catalogue	2020	S,C	25	S2	513x13	1022	20m	15.4	?	?	?	—
78	SAR-based Sea Ice Dataset	2020	C	6	SAR (Satellite)	98.8K	28	—	0.14	?	?	?	—
79	SenseEarth classify	2020	C	51	RGB	70K	100~12655	0.2~153m	10.8	?	?	?	—
80	Urban Tree Detection Data	2020	C	2	RGB-NIR	60	256	0.6m	0.114	?	?	?	—
81	2019 Mali CropType	2021	C	4	MS (Satellite)	33*13*41	256	—	0.456	?	?	?	—
82	CropHarvest	2021	C	—	Multi	95.2K	—	10~60m	22.7	Yes	?	?	TG
83	CropRotation	2021	C	—	MS (Satellite)	—	—	—	53.4	?	?	?	—
84	DENETHOR	2021	C	9	S1+S2	2*(365 + 60*14)	8000	3m	254	?	?	?	—
85	FGSCR-42	2021	C	42	RGB	9.3K	500	—	—	?	?	?	—
86	Forest Damages – Larch Case-bearer	2021	OD,C	4	RGB	1.5K	1500	—	3.3	?	?	?	—

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
87	MAFAT-Fine-Grained	2021	C	37	RGB	4.2K	—	0.05m-0.15m	—	?	?	—	
88	Million AID	2021	C	28	RGB	1M	150~550	0.5~153m	134	Yes	?	Paper	
89	MultiScene	2021	C	36	RGB	100K	512	0.3~0.6m	6	Yes	?	Paper	
90	NaSC-TG2	2021	C	10	RGB-NIR	20K	256	100m	—	?	?	—	
91	Satellite Image Classification	2021	C	4	RGB	5.6K	256	—	0.023	?	?	—	
92	SeaIceClassification	2021	C	8	SAR (Satellite)	6.4K	256	—	2.9	?	?	—	
93	South Africa Crop Type	2021	C	9	S1+S2	122.7K	—	—	82.8	Yes	?	TG	
94	TG1HRSSC	2021	C	9	HS	204	512	5m-20m	0.277	?	?	—	
95	Urban Village S2UC	2021	C	2	RGB	1.7K	224	2m	1.8	?	?	—	
96	ZueriCrop	2021	C	48	S2	28K×71	24	10m	38.5	Yes	?	TG	
97	ADVANCE	2022	C	13	OPT (Satellite)	5.1K	256	—	4.4	Yes	?	TG	
98	AgriFieldNet	2022	C	13	MS (Satellite)	1.2K×12	256	—	0.32	Yes	?	TG	
99	Aircraft SAR-ACD	2022	C	20	SAR	4.3K	64	—	—	?	?	—	
100	AnthroProtect	2022	C	2	OPT (Satellite)	23.9K	256	—	18.1	?	?	—	
101	ARGO	2022	C	2	MS (Satellite)	1.8K	1400	—	14.3	?	?	—	
102	CaiRoad	2022	C	—	Satellite MT	—	—	—	10.7	Yes	Yes	GFM+Paper	
103	CalFire	2022	C	—	Satellite MT	—	—	—	7.2	Yes	Yes	GFM+Paper	
104	Canadian Cropland	2022	C	10	S2	78.5K	64	10m	26	Yes	Yes	GFM	
105	fMoW-Sentinel	2022	C	62	S2 MS	882.8K	96	10~60m	—	Yes	Yes	GFM	
106	MUSIC4HA (Hot Area)	2022	C	6	S2	2.5K	16	10m	0.01	?	?	—	
107	OMBRIA	2022	C	2	S1+S2	3.4K	256	10m-20m	0.19	?	?	—	
108	ReforesTree	2022	OD,C	6	OPT (Drone)	100	4000	—	7.5	Yes	Yes	FOMO+TG	
109	S2-Agri	2022	C	—	MS (Satellite)	—	32	—	10.8	?	?	—	
110	Space2Ground	2022	C	2	Multi	10.1K	260	10~60m	0.501	?	?	—	
111	SynthWakeSAR	2022	C	10	SAR	46.1K	96000	3.3m	4.3	?	?	—	
112	The Auto Arborist Dataset	2022	C	344	RGB+MS	2.6M	1~24	—	24	Yes	?	Paper	
113	TreeSatAI	2022	C	47	Multi	50.4K	304	0~1	16.3	Yes	Yes	FOMO+GB2+GFM+Pa	
114	UAE-RS	2022	C	—	OPT	6.1K	—	—	—	?	?	—	
115	XAI4SAR	2022	C	—	SAR (Satellite)	5.9K	256	—	0.691	?	?	—	
116	DFC23	2023	S,C	12	Multiple (Satellite)	—	—	—	—	?	?	—	
117	Digital Typhoon	2023	C,R	8	IR	189.4K	512	5000m	—	Yes	?	TG	
118	FireRisk	2023	C	7	OPT (Satellite)	91.9K	—	—	14	Yes	?	TG	
119	MAR20	2023	C	20	RGB	3.8K	800×800	1m	1.1	Yes	?	Paper	
120	m-BigEarthNet (GEO-Bench)	2023	C	43	S2 MS	22K	120	10m	—	Yes	Yes	GB1	
121	m-Brick-Kiln (GEO-Bench)	2023	C	2	S2 MS	17,061	64	10m	1.6	Yes	Yes	GB1	
122	m-EuroSAT (GEO-Bench)	2023	C	10	S2 MS	4K	64	10m	1.2	Yes	Yes	GB1	
123	m-ForestNet (GEO-Bench)	2023	C	12	Landsat-8 MS	8,446	332	15m	1.2	Yes	Yes	GB1	
124	m-PV4GER (GEO-Bench)	2023	C	2	RGB	13,812	320	0.1m	0.68	Yes	Yes	GB1	
125	m-So2Sat (GEO-Bench)	2023	C	17	S1/S2	21,964	32	10m	—	Yes	Yes	GB1	
126	BigEarthNet v2.0 (S1+S2)	2024	C,S	19	S1/S2	549.5K pairs	120	10~60m	110	Yes	Yes	GB2+Paper	
127	EuroCropsML	2024	C	176	S2 MS	706.7K	—	10m	—	?	?	—	
128	M3LEO	2024	P,C	—	S1/S2	10M	—	10m	17.5K	No	No	—	
129	PASTIS-HD	2024	S,C,I	20	Multi	—	—	1m	—	Yes	Yes	GFM	
130	PhilEO Bench	2024	R,S,C	—	S2 MS	—	—	10m	400	Yes	Yes	PH	
131	PhilEO-downstream	2024	S,C,R	11	S2	—	256	10m	400	?	?	—	
132	QuakeSet	2024	C,R	2	S1 SAR	3.3K	512	10m	11	Yes	?	TG	
133	reBEN	2024	S,C	—	S1/S2	549.5K	20~120	10m-60m	—	?	?	—	
134	SICKLE	2024	S,C	21	S1/S2+L8 TS	34.8K	—	3m-30m	—	Yes	Yes	GFM	
135	TalloS	2024	C	1160	Multi	335K	—	10m	—	Yes	Yes	FOMO	
136	TreeSatAI-TS	2024	C	20	Multi	—	—	0.2m-10m	30	?	?	—	
137	BigEarthNet-L (Landsat-Bench)	2025	C	19	Landsat MS	—	—	30m	—	Yes	Yes	LB	
138	BigEarthNetV2-S2 (Copernicus-Bench)	2025	C	19	S2 MS	24K	120	10m	—	Yes	Yes	CB	
139	EuroSAT-L (Landsat-Bench)	2025	C	10	Landsat MS	—	—	30m	—	Yes	Yes	LB	
140	GEO-Bench-2	2025	OD,I	—	Multi	19 datasets	—	—	—	Yes	Yes	GB2	
141	Landsat-Bench	2025	C	—	Landsat MS	3 datasets	—	30m	—	Yes	Yes	LB	
142	LC100-L (Landsat-Bench)	2025	C	—	Landsat MS	—	—	30m	—	Yes	Yes	LB	
143	MONITRS	2025	C,IC	—	S2 RGB+Text	10K	512	10m	6.39	?	?	—	
144	REOBench	2025	OD,IC	—	OPT+Text	6 subsets	—	—	—	Yes	Yes	REO	
145	CLearS	2026	C,OD	—	Multi	7 datasets	—	—	—	?	?	—	
146	CropClimateX	2026	R,C,T	—	Multi	15.5K cubes	—	—	6.3K	?	?	—	
147	SwissCrop25	2026	C,T	73	S2 MS	7 seasons	—	10m	—	?	?	—	
148	Earth Surface Temperature	2017	R	—	Climate variables	16	—	—	0.09	?	?	—	
149	Greenhouse Gas	2017	R	—	Gas emission	—	—	—	0	?	?	—	
150	Forest Canopy Height	2018	R	—	LiDAR	1.1K	—	1m	62.9	?	?	—	
151	TenGeoP-SARwv	2019	R	10	SAR	37K	—	5m	31.7	?	?	—	
152	Poverty in Africa	2020	R	—	RGB	32.8K	256	—	5.58	?	?	—	
153	Tropical Cyclone	2020	R	—	GOES 8–16 MSI	108.1K	256×256	4K–8K	—	Yes	?	TG	
154	Air Quality e-Reporting	2021	R	—	—	—	—	—	—	?	?	—	
155	Forel-Ule Index global inland waters	2021	R	—	MS	—	—	500m	0.004	?	?	—	
156	Hurricane Wind Speed	2021	R	—	MS	114.6K	—	—	—	?	?	—	
157	SustainBench Crop Yield	2021	R	—	MODIS MSI	11K	32	—	—	Yes	?	TG	
158	USAVars	2021	R	—	NAIP RGB+NIR	100K	—	4m	—	Yes	?	TG	
159	Hyperview Challenge	2022	R	—	HS	2.9K	11~250	2m	1.85	Yes	?	Name+Paper	
160	So2Sat Population	2022	R	—	DEM+S2	—	—	10m	—	Yes	?	Paper	
161	BioMassters	2023	R	—	S1/S2	13K	256	10m	201	Yes	Yes	GB2+PG+TG	
162	SKIPP'D	2023	R	—	RGB	363.4K	64	—	—	Yes	?	TG	
163	Open-Canopy	2024	R,CD	—	SPOT-6/7+LiDAR	>87K km ²	1K tiles	1.5m	—	Yes	Yes	PG+Paper	
164	SolarCube	2024	R,T	—	Multi	—	—	—	—	?	?	—	
165	Western USA Live Fuel Moisture	2024	R	—	L8+S1	2.6K	—	—	—	Yes	?	TG	
166	YieldSAT	2026	R,T	—	Multi	12.2M	—	10m	—	?	?	—	
167	Air Quality	—	R	—	Multi-sensor	9.4K	—	—	—	Yes	?	TG	

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Table 1 continued

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168	DFC07	2007	S	–	Multiple (Satellite)	1	787	–	0.019	?	?	–	
169	DFC08	2008	S	5	HS (Aerial)	–	–	–	0.194	?	?	–	
170	ISPRS 2D - Potsdam	2011	S	6	RGB+nDSM	38	6000	0.05m	16	Yes	?	Paper+TG	
171	ISPRS 2D - Vaihingen	2011	S	6	RGB+nDSM	33	2200	0.09m	17	Yes	Yes	GFM+Paper+TG	
172	Aerial Image Segmentation	2013	S	–	RGB	80	512	0.3~1m	0.007	?	?	–	
173	DFC13	2013	S	16	Multiple (Aerial)	145	–	–	1	?	?	–	
174	Massachusetts Roads	2013	S	–	RGB	1.2K	1500	1m	7.55	?	?	–	
175	DFC14	2014	S	7	Multiple (Aerial)	–	–	–	–	?	?	–	
176	DFC15	2015	S	–	Multiple (Aerial)	7	10000	–	–	?	?	–	
177	DFC2015 Zeebruges	2015	S	8	Multi	7	0~100	0.05m	0.002	Yes	?	Paper	
178	ERM PAIW	2015	S	2	RGB	41	4000	0.3m	0.635	?	?	–	
179	Zurich Summer	2015	S	–	RGB-NIR	20	1000	0.61m	0.132	?	?	–	
180	Chikusei	2016	S	19	HS (Aerial)	1×128	2517	–	1.4	?	?	–	
181	DFC16	2016	S	–	OPT	–	–	–	–	?	?	–	
182	HD-Maps	2016	S	5	RGB	20	4000	0.3m	0.146	?	?	–	
183	L8 Biome	2016	S	5	L8 MSI	96	8.9K	15~30m	96	Yes	?	TG	
184	SpaceNet-1	2016	S	1	RGB+MS	9.7K	650	0.5~1m	31	Yes	?	Paper	
185	AIS From Online Maps	2017	S	2	RGB	1.7K	3000	0.5m	23.8	?	?	–	
186	CITY-OSM	2017	S	–	OPT (Aerial)	21	2500	–	23.8	?	?	–	
187	DFC17	2017	S	17	Multiple (Multiple)	–	–	–	–	?	?	–	
188	EvLab-SS	2017	S	11	RGB	60	4500	0~0.2	–	?	?	–	
189	GTA_BEV_SVA	2017	S	–	OPT	1M	–	–	–	?	?	–	
190	INRIA	2017	S	1	RGB	360	1500	0.3m	19.5	Yes	?	Paper	
191	Inria Aerial Image Labeling	2017	S	2	RGB	360	5000	0.3m	–	Yes	?	Paper+TG	
192	SpaceNet-2	2017	S	1	RGB+MS	24.6K	650	0.3m	182	Yes	Yes	GB2+GFM+Paper	
193	SpaceNet-3	2017	S	1	Multi	3.7K	1300	0.3~1.24m	182	Yes	?	Paper	
194	Urban 3D Challenge	2017	S	2	RGB	157K	–	0.5m	–	Yes	?	Name	
195	38-Cloud	2018	S	1	RGB	17.6K	384	30m	16	?	?	–	
196	AerialLanes18	2018	S	–	RGB	20	5616	0.125m	–	?	?	–	
197	Aeroscapes	2018	S	11	RGB	3.3K	1280	–	0.752	?	?	–	
198	Agricultural Crop Cover	2018	S	2	MS	40	–	30m	4.4	?	?	–	
199	DeepGlobe (Building)	2018	I,S	2	Multi	24.6K	650	0.5m	–	?	?	–	
200	DeepGlobe-LandCover	2018	S	7	RGB	1.1K	2448	0.5m	2.75	Yes	?	TG	
201	DeepGlobe-Road	2018	S	1	OPT (Satellite)	8.6K	1024	–	3.84	Yes	?	Paper	
202	DFC18	2018	S	20	Multi	1	–	0.05~1m	19.8	Yes	?	Paper	
203	DLRSD	2018	S	21	RGB	2.1K	256	0.3m	0.004	Yes	?	TG	
204	GID-15 (Gaofen Image Dataset)	2018	S	15	GF-2 RGB	150	6800×7200	3m	36	Yes	?	Paper+TG	
205	HyRANK	2018	S	14	HS	5	0~1	30m	0.4	?	?	–	
206	RIT-18	2018	S	–	MS	1	9300	0.047m	1.5	?	?	–	
207	RoadNet	2018	S	2	RGB	20	–	0.21m	0.905	?	?	–	
208	SEN1-2	2018	S	–	Multiple (Satellite)	282.4K	256	–	43.7	?	?	–	
209	SynthAer	2018	S	8	RGB	765	1280	–	1	?	?	–	
210	Urban Drone Dataset (UDD)	2018	S	6	RGB	301	4~96	–	1.1	?	?	–	
211	WHDL	2018	S	6	RGB	4.9K	256	2m	0.102	?	?	–	
212	AeroRIT	2019	S	–	RGB+HS	1	3900	0.4m	1.8	?	?	–	
213	AIRS	2019	S	1	RGB	1K	10000	0.075m	17.6	?	?	–	
214	Chesapeake Land Cover	2019	S	16	RGB+MS	250K	240	1m	140	Yes	?	TG	
215	DLR-SkyScapes	2019	S	31	RGB	16	4680	0.13m	–	?	?	–	
216	DroneDeploy	2019	S	7	RGB	55	0~6	0.1m	–	?	?	–	
217	GF2 Dataset for 3DFGC	2019	S	5	RGB-NIR	11	2~652	4m	0.056	?	?	–	
218	HRC_WHU	2019	S	2	RGB	150	1280×720	0.5~15m	0.17	?	?	–	
219	ManipalUAVid	2019	S	4	OPT (Drone)	711	1280	–	1.05	?	?	–	
220	Open AI Challenge: Caribbean	2019	S	5	RGB	7	52~318	0.04m	–	Yes	?	Name+Paper	
221	PolSF	2019	S	6	SAR (Satellite)	5	–	–	–	?	?	–	
222	Salient Objects ORSSD	2019	S	8	RGB	800	500	–	0.026	?	?	–	
223	Semantic Drone	2019	S	20	RGB	400	5000	–	4	?	?	–	
224	SpaceNet-5	2019	S	1	Multi	2.4K	1300	0.3m	84.1	Yes	?	Paper	
225	95-Cloud	2020	S	1	RGB	34.7K	384	30m	18	?	?	–	
226	ActiveFireL8	2020	S	1	MS (Satellite)	2M×10	128	–	546	?	?	–	
227	AIR-CD	2020	S	1	RGB-NIR	34	7300	4m	12	?	?	–	
228	BDCI2020	2020	S	7	RGB	146K	256	2m	1.3	Yes	?	Paper	
229	BHPools&WaterTanks	2020	S	1	OPT (Aerial)	350	3000	–	1.9	?	?	–	
230	DFC20	2020	S	17	S1+S2	180.7K	–	10m	19.2	Yes	?	Paper	
231	DFC2020	2020	S	10	S1/S2	5.1K	256	10m	–	Yes	Yes	CB+GFM	
232	DubaiSemSeg	2020	S	6	OPT (Satellite)	72	800	–	0.032	?	?	–	
233	EORSSD	2020	S	2	RGB	2K	500	–	0.06	?	?	–	
234	FUSAR-Map	2020	S	4	SAR (Satellite)	610	1024	–	–	?	?	–	
235	Ghana Crop Type	2020	S	5	Multi	2881*14*25	64	3~10m	312	?	?	–	
236	GTA-V SID	2020	S	–	OPT (Satellite)	121	500	–	0.1	?	?	–	
237	HKH Glacier Mapping	2020	S	2	MS (Satellite)	14.2K×15	512	–	29.4	?	?	–	
238	Kaggle aerial segmentation	2020	S	6	RGB	72	800	–	0.033	?	?	–	
239	Kaggle Buildings	2020	S	2	OPT (Satellite)	6K	256	–	0.878	?	?	–	
240	Kaggle Water Bodies	2020	S	2	MS (Satellite)	2.8K	1000	–	0.274	?	?	–	
241	Kag. Massachusetts Buildings	2020	S	1	RGB	151	1500	1m	3	?	?	–	
242	Kenya Crop Type	2020	S	7	S2 MSI	4.7K	3035×2016	10m	–	Yes	?	TG	
243	LandCoverAI	2020	S	–	RGB	41	9000	0~0.2	1.4	Yes	?	TG	
244	Open Cities AI Challenge	2020	S	2	RGB	11K	1~24	0.03~0.2m	81.5	Yes	?	Name+Paper	
245	OpenSARUrban	2020	S	10	SAR (Satellite)	33.4K	100	–	2.2	?	?	–	
246	Parking APKLOT	2020	S	2	RGB	501	–	–	3	?	?	–	
247	Power TTPL	2020	S	3	RGB	1.1K	3840	–	4.2	?	?	–	
248	Rwanda Built-Up	2020	S	1	OPT (Satellite)	787	256	–	0.07	?	?	–	
249	SemCity Toulouse	2020	S	–	MS	16	3500	0.5~2m	5.2	Yes	?	Paper	
250	SEN-12-FLOOD	2020	S	1	Multi	336	–	10m	12.2	?	?	–	
251	Sen1Floods11	2020	S	2	S1/S2	4.8K	512	10m	14.3	Yes	Yes	GFM+PG	
252	South Sudan Crop Type	2020	S	5	S1/S2+Planet	837 tiles	64	10m	977	Yes	Yes	PG	
253	SpaceNet-6	2020	S	1	SAR+RGB	3.4K	900	0.5m	0.368	Yes	Yes	GB2	
254	SpaceNet-7	2020	CD,S	1	MS+MT	1.5K	1024	4m	20.6	Yes	Yes	GB2+PG	
255	Synthinel-1	2020	S	1	RGB	2.1K	572	0.3m	1	?	?	–	
256	TimeSen2Crop	2020	S	16	S2	1M	10980	10m	1.1	Yes	Yes	GFM	
257	TTPLA	2020	S	–	OPT (Aerial)	1.1K	3840	–	4.2	?	?	–	

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
258	UAVid	2020	S	–	RGB	420	4000	–	5.88	?	?	–	
259	VALID	2020	OD,S	30	RGBD	6.7K	1024	–	15.7	?	?	–	
260	WHU Cloud	2020	OD,S	1	RGB	859	512	30m	3.56	?	?	–	
261	WHU-Hi	2020	S	11	HS (Drone)	3	1217	0.109m	0.817	Yes	?	Paper	
262	WHU-Hi-HongHu	2020	S	22	HS	1	940x475	0.043m	–	Yes	?	Paper	
263	Arctic Sea Ice Masking	2021	S	8	RG-NIR	3.4K	357x306	10m	0.092	?	?	–	
264	Benin Cashews	2021	S	6	S2 (labels corrected with Pléiades)	70	1186x1122	0~1	–	Yes	?	TG	
265	DFC21-DSE	2021	S	4	Multi	6K	16	10~750m	18	Yes	?	Paper	
266	DoriaNet	2021	S	3	OPT (Drone)	2.4K	1280	–	0.067	?	?	–	
267	ETCI2021	2021	S	2	SAR (Satellite)	66.8K	256	–	5.3	Yes	?	TG	
268	EuroCrops	2021	S	43	S2	805.4K	–	10m	8.6	?	?	–	
269	FloatingObjects	2021	S	–	MS (Satellite)	–	–	–	15.4	?	?	–	
270	FloodNet	2021	S	10	RGB	2.3K	4000	0.015m	12.1	?	?	–	
271	LoveDA	2021	S	7	RGB	6K	1024	0.3m	9.6	Yes	?	TG	
272	MARIDA	2021	S	15	MS (Satellite)	1.4K×10	256	–	1.2	Yes	Yes	GFM	
273	Mini Inria Aerial Image Labeling Dataset	2021	S	2	RGB	32.5K	512	0.3m	–	Yes	?	Paper	
274	MUESLI	2021	S	41	Multiple (Aerial)	1×415	12800	–	37.7	?	?	–	
275	MultimodalBen	2021	S	30	Multiple (Multiple)	568	680	–	0.489	?	?	–	
276	PASTIS	2021	S	18	S2	2433*45*10	128	10m	29	Yes	Yes	GB2+TG	
277	PASTIS-R	2021	S	18	Multiple (Satellite)	2.4K	128	–	54	Yes	Yes	PG	
278	RescueNet	2021	S	12	OPT (Drone)	4.5K	3500	–	21.8	?	?	–	
279	Rwanda Field Boundary	2021	S	2	RGB+NIR	70	256	4.7m	–	Yes	?	TG	
280	S2-SHIPS	2021	S	3	MS (Satellite)	16×16	1783	–	2.9	?	?	–	
281	Sen4AgriNet	2021	S	–	S2	–	–	10~60m	262	Yes	?	Paper	
282	SmallholderPlant	2021	S	6	MS (Satellite)	70×13	1180	–	1	?	?	–	
283	The Azavea Cloud Dataset	2021	S	2	S2	32	–	10m-60m	–	?	?	–	
284	UOPNOA	2021	S	–	Multiple (Multiple)	36K	–	–	9.7	?	?	–	
285	AIR-PolSAR-Seg	2022	S	6	SAR	2K	512	8m	0.609	?	?	–	
286	Burned Area Delineation	2022	S	–	Multiple (Satellite)	–	–	–	14	?	?	–	
287	C2S-MS	2022	S	3	Multiple (Satellite)	1.8K	512	–	12	?	?	–	
288	CloudSEN12	2022	S	4	S2/S1+DEM	49.4K	509	10m	~1K	Yes	Yes	CB+GB2+Paper	
289	DFC2022	2022	S	15	RGB	4K	2000	0.5m	–	Yes	?	TG	
290	DFC22-SSL	2022	S	14	Multiple (Aerial)	5K	2000	–	42	?	?	–	
291	DynamicEarthNet	2022	CD,S	7	PlanetFusion	54.8K	1024	3m	524	Yes	Yes	GB2+GFM+PG	
292	FireCube	2022	S	1	Multiple (Multiple)	4.3K×58	700	–	23.5	?	?	–	
293	Five-Billion-Pixels	2022	S	24	RGB+MS	150	7200×6800	4m	104	Yes	Yes	PG	
294	FLAIR #1	2022	S	13	OPT (Aerial)	77.4K	512	–	70	?	?	–	
295	GeoNRW	2022	S	10	RGB+nDSM	7.8K×3	1000	1m	30	Yes	?	TG	
296	GrowliFlower	2022	S	4	MS (Drone)	8.4K	256	–	11.7	?	?	–	
297	HOSD	2022	S	1	HS (Aerial)	18×224	1700	–	2.2	?	?	–	
298	Landslide4Sense	2022	S	1	Multiple (Satellite)	4.8K×14	128	–	2.4	?	?	–	
299	MapInWild	2022	S	11	Multiple (Satellite)	1K×8	1920	–	365	Yes	?	TG	
300	MDAS	2022	S	20	Multi	3	100~18K	0.3~30m	–	Yes	?	TG	
301	Microsoft RoadDetections	2022	S	1	RGB	20K	1088	1m	9.25	?	?	–	
302	MiniFrance	2022	S	12	RGB	2.1K	10000	0.5m	93.1	?	?	–	
303	MMFlood	2022	S	1	SAR (Satellite)	1.7K×2	2000	–	11.3	Yes	?	TG	
304	MultiSenGE	2022	S	14	S1+S2	1.1M×12	256	10m	530	Yes	?	Paper	
305	NASA-Harvest	2022	S	1	MS (Satellite)	70	256	–	0.3	?	?	–	
306	NDWS	2022	S	1	Multiple (Multiple)	18.4K×12	64	–	2	?	?	–	
307	OnCloudN	2022	S	1	OPT (Satellite)	22.7K	512	–	50	?	?	–	
308	OpenEarthMap	2022	S	8	RGB	5K	1024	0.25m-0.5m	9.1	?	?	–	
309	OpenSentinelMap	2022	S	15	RGB+S2	137K	192	10m-60m	445	?	?	–	
310	Paddy Rice Labeling South Korea	2022	S	–	S2	–	–	–	0.002	?	?	–	
311	Paddy Rice Maps South Korea	2022	S	–	S1	12.9K	256	10m	0.198	?	?	–	
312	RDCME	2022	S	1	OPT (Satellite)	775	256	–	–	?	?	–	
313	RID	2022	S	16	RGB	2K	–	0.1m	1.5	?	?	–	
314	S12 ScotInd Cloud	2022	S	1	Multiple (Satellite)	–	–	–	0.96	?	?	–	
315	Sat Burn Area	2022	S	1	MS (Satellite)	73×14	1500	–	14	?	?	–	
316	SeasoNet	2022	S	33	S2	1.8M	120	10m	229	Yes	?	TG	
317	Sentinel-2 Cloud Cover	2022	S	2	MS	22.7K	–	10m-60m	51.2	Yes	?	TG	
318	SpaceNet-8	2022	S	2	PAN+RGB	1.2K×2	1200	0.3~0.8m	6.8	Yes	Yes	GB2	
319	SyntCities	2022	S	13	OPT	16.2K	–	–	–	?	?	–	
320	TimeSpec4LULC	2022	S	29	MS+MT	–	–	500m	60	?	?	–	
321	Urban	2022	S	61	OPT (Satellite)	800	600	–	0.675	?	?	–	
322	Urban Building Classification	2022	S	61	RGB	800	600	0.5~0.8m	0.675	?	?	–	
323	WHU-OHS	2022	S	24	HS	7.8K	512	10m	94.9	Yes	?	Paper	
324	WildfireDanger	2022	S	1	Multiple (Multiple)	–	–	–	17	?	?	–	
325	AI4SmallFarms	2023	S	2	S2 MS	62 tiles	–	10m	–	Yes	Yes	PG	
326	BraDD-S1TS	2023	CD,S	2	S1 TS	13.2K	–	10m	17.7	Yes	Yes	GFM	
327	DW-Expert	2023	S	9	S1/S2	71.4K	96	10m	–	Yes	Yes	GFM	
328	FLAIR #2	2023	S	13	Multi	77.8K	512	0.2m-10m	–	Yes	Yes	FOMO+GB2+GFM	
329	HLS Burn Scars	2023	S	2	HLS	804	512	30m	–	Yes	Yes	GB2+GFM+PG	
330	I/O Bench	2023	S	–	Landsat MSI	1	8000	30m	–	Yes	?	TG	
331	LuoJia-HSSR	2023	S	23	RGB	6.4K	256	0.75m	–	?	?	–	
332	m-Cashew-Plantation (GEO-Bench)	2023	S	7	S2 MS	1,800	256	10m	1.3	Yes	Yes	GB1	
333	m-Chesapeake-LandCover (GEO-Bench)	2023	S	7	RGBN	5K	256	1m	–	Yes	Yes	GB1	
334	m-NeonTree (GEO-Bench)	2023	S	2	RGB+HS+DEM	457	400	0.1m	–	Yes	Yes	GB1	
335	m-NZ-Cattle (GEO-Bench)	2023	S	2	RGB	655	500	0.1m	0.33	Yes	Yes	GB1	
336	m-PV4GER-Seg (GEO-Bench)	2023	S	2	RGB	3,806	320	0.1m	0.32	Yes	Yes	GB1	
337	m-SA-Crop-Type (GEO-Bench)	2023	S	10	S2 MS	5K	256	10m	1.4	Yes	Yes	GB1	
338	SSL4EO-L Benchmark	2023	S	17/134	Landsat MSI	25K	264	30m	–	Yes	?	Name+TG	
339	Chesapeake Roads	2024	S	1	RGBN	–	–	1m	28.3	?	?	–	
340	Fields Of The World	2024	S,I	2/3	S2 MSI	70.8K	256	10m	–	Yes	Yes	GB2+TG	
341	FUSU	2024	S,CD	17	S1/S2	>30B px	–	10m	–	?	?	–	
342	GAMUS	2024	S	7	RGB+nDSM	–	–	1m	10	Yes	?	Paper	
343	Kuro Siwo	2024	S	3	S1 SAR+DEM	67.5K labeled	224	10m	–	Yes	Yes	CB+GB2+Paper	

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	EvalEvidence
344	MADOS	2024	S	15	S2 MS	2.8K	40~240	10~60m	5.35	Yes	Yes	PG
345	QuickQuakeBuildings	2024	S	2	RGB+SAR	—	—	0.3m	—	?	?	—
346	Ticino	2024	S	10	Multi	—	—	1.86m-30m	—	?	?	—
347	UrbanSARFloods	2024	S	2	S1 SAR	8.9K	—	10m	—	?	?	—
348	WALD5Dplus	2024	S	—	RGB+SAR	—	—	10m	—	?	?	—
349	DL4GAM	2025	S	2	S2 MSI	2.3K/11.4K	256	10m	—	Yes	?	TG
350	OpenEarthMap-SAR	2025	S	8	SAR+OPT	1.5M seg-ments	—	—	—	?	?	—
351	50 Cities	2026	S	—	S1/S2	50	—	10m	10.7	?	?	—
352	BELDE	2026	S	7	S2 MS	1.1M	—	10m	—	?	?	—
353	CropSeg	2026	S	—	HLS MS	3,854	—	30m	—	Yes	Yes	GFM
354	FLAIR-HUB	2026	S	19/23	Multi	241.1K	10~512	0.2~10m	—	Yes	?	TG
355	CaFFe	—	S	2/4	SAR	19.1K	512	6~20m	—	Yes	Yes	GB2+TG
356	EnviroAtlas	—	S	10	Multi	—	—	1m	—	Yes	?	TG
357	L7 Irish	—	S	5	L7 MSI	206	7.5K~8.4K	15~30m	—	Yes	?	TG
358	Solar Plants Brazil	—	S	2	RGB+NIR	272	256	10m	—	Yes	?	TG
359	Substation	—	S	2	S2 MSI	27K	228	10m	—	Yes	Yes	GB2+TG
360	CrowdAI Mapping	2018	I	1	RGB	401.8K	300	—	5.33	?	?	—
361	iSAID	2020	I	15	RGB	2.8K	4000	—	6.54	?	?	—
362	GlobalMLBuildingFootprints	2024	I	2	Aerial RGB	—	—	0.5m	—	?	?	—
363	SEN12TP	2023	T	—	S1/S2	2.3K	—	—	—	?	?	—
364	DFC09	2009	CD	—	Multiple (Satellite)	2	—	—	0.098	?	?	—
365	SZTAKI AirChange	2012	CD	2	RGB	13	800	1.5m	0.042	?	?	—
366	Cross-sensor Bastrop	2015	CD	2	MS	4	0~153	3~12	—	?	?	—
367	AIST Building Change Detection	2017	CD	2	RGB	16.9K	160	0.4m	17.8	?	?	—
368	HSI CD	2017	CD	5	HS (Multiple)	3×224	800	—	1.7	?	?	—
369	AICD	2018	CD	—	OPT (Aerial)	1K	800	—	1.7	?	?	—
370	CDD (season-varying)	2018	CD	2	RGB	16K	256	0.03~0.1m	2.7	?	?	—
371	GETNET dataset	2018	CD	2	HS	1	463×241	30m	0.05	?	?	—
372	Hermiston City Oregon	2018	CD	5	HS	1	390×200	30m	—	?	?	—
373	Hyperspectral CD	2018	CD	5	HS	3	0~98	30m	1.7	?	?	—
374	OSCD	2018	CD	2	S2	24	600	10m	0.48	Yes	Yes	GFM+TG
375	Synthetic & Real	2018	CD	—	OPT (Aerial)	16K	256	—	2.7	?	?	—
376	WHU Building change detection	2018	CD	2	RGB	2	15354×32507	0.075m	5	?	?	—
377	xView2	2018	CD	4	RGB	22.1K	1024	0.5m	51	Yes	Yes	PG+TG
378	Flood California flood dataset	2019	CD	2	RGB+MS	1	1534×808	5m,30m	0.33	?	?	—
379	HRSCD	2019	CD	7	RGB	582	10000	0.5m	5.06	?	?	—
380	MtS-WH	2019	CD	1	RGB-NIR	190	150	1m	0.459	?	?	—
381	DSIFN	2020	CD	1	RGB	394	512	10m	0.454	?	?	—
382	Google Data Set	2020	CD	2	RGB	19	1000~5000	0.55m	0.6	?	?	—
383	Hi-UCD	2020	CD	9	RGB	1.3K	1024	0.1m	—	?	?	—
384	LEVIR-CD	2020	CD	1	RGB	637	1024	0.5m	2.7	Yes	Yes	GFM+TG
385	SECOND	2020	CD	30	RGB	4.7K	512	0.5~3m	2.2	?	?	—
386	Zhang et al. CD dataset	2020	CD	2	RGB+NIR	4	458~1.4K	2~5	0.1	?	?	—
387	DFC21-MSD	2021	CD	15	MS+MT	2.2K×9	4000	1~30m	325	Yes	?	Paper
388	Dynamic EarthNet Challenge	2021	CD	7	RGB	22.5K	1024	3m	—	Yes	?	Name
389	HTCD	2021	CD	2	RGB	3.8K	256~2K	0.074m-0.597m	1.74	?	?	—
390	MultimodalSupCD	2021	CD	—	Multi	—	—	—	—	?	?	—
391	MUNO21	2021	CD	1	OPT (Aerial)	156	12000	—	14.3	?	?	—
392	RapidAI4EO	2021	CD	44	MS+RGB	500K	—	3m,10m	—	Yes	Yes	FOMO
393	Relative Radiometric Normalization	2021	CD	2	MS	7	300~5000	0~6	1	?	?	—
394	S2Looking	2021	CD	1	RGB	5K×2	1024	0.5~0.8m	10.2	?	?	—
395	S2MTCP	2021	CD	—	S2	1.5K	600	10m	10.6	?	?	—
396	SYSU-CD	2021	CD	5	RGB	20K×2	256	0.5m	5.1	?	?	—
397	WH-MAVS	2021	CD	15	RGB	47.1K	200	1.2m	—	Yes	?	Paper
398	CDVQA	2022	CD	19	RGB	3K	512	0.5~3m	—	?	?	—
399	Cropland Change Detection (CLCD)	2022	CD	2	RGB	600	512	0.5m-2m	0.5	?	?	—
400	Dynamic World	2022	CD	9	S2	—	—	10m	—	?	?	—
401	MSBC	2022	CD	2	Multi	3.8K	256	2m	3.9	?	?	—
402	MSOSCD	2022	CD	2	Multi	5.1K	256	10~60m	2.7	?	?	—
403	BANDON	2023	CD	2	RGB	2.3K	2048×2048	0.6m	3.4	?	?	—
404	ChangeNet	2024	CD	6	Multi-temporal RGB	—	—	0.3m	—	?	?	—
405	BRIGHT	2025	CD	4	RGB+SAR	3.2K	1024	0.1~1m	—	Yes	?	TG
406	FAIR-EO CD Benchmark	2026	CD	—	Multi	—	—	—	—	Yes	?	Name
407	CaBuAr	—	CD	2	S2 MSI	424	512	20m	—	Yes	?	TG
408	ChaBuD	—	CD	2	S2 MSI	356	512	10m	—	Yes	?	TG
409	MSTAR	1996	OD	—	SAR (Satellite)	—	—	0.3m	—	?	?	—
410	SAR Vehicle	2005	OD	1	SAR (Aerial)	104	—	—	7.5	?	?	—
411	VIVID - PETS2005	2007	OD	1	OPT (Aerial)	9.3K	640	—	—	?	?	—
412	TaS	2008	OD	1	RGB	30	792	0.5m	0.97	?	?	—
413	OIRDS	2009	OD	5	RGB	900	256~640	0.15m	0.153	?	?	—
414	NWPU-VHR10	2014	OD	10	RGB+IRRG	800	1000	0.08~2m	0.073	Yes	?	TG
415	UCAS_AOD	2014	OD	2	RGB	976	0~1	—	3.24	?	?	—
416	PKLot	2015	OD	1	RGB	12.4K	1280	—	4.6	?	?	—
417	CARPK	2016	OD	1	RGB	1.4K	1280	—	2	?	?	—
418	COWC	2016	OD	1	RGB	388.4K	256	0.15m	64	Yes	?	TG
419	DLR3k/DLR-MVDA	2016	OD	7	RGB	20	3744	0.13m	0.162	?	?	—
420	DSTL16B	2016	OD	10	MS (Satellite)	1.4K	3391	—	7.3	?	?	—
421	DSTL3B	2016	OD	10	OPT (Satellite)	450	3391	—	12.9	?	?	—
422	HSI Vehicle	2016	OD	1	HS (Aerial)	157	—	—	—	?	?	—
423	LEVIR	2016	OD	3	RGB	22K	800	0.5m	—	?	?	—
424	Stanford Drone Dataset	2016	OD	6	RGB Video	—	—	0.025m	69	?	?	—
425	HRSC2016	2017	OD	26	RGB	1.1K	1100	0.4~2m	8	?	?	—
426	Land Covers Dstl Satellite Imagery	2017	OD	10	RGB+MS	57	3~348	0.3m-7.5m	21.7	?	?	—
427	OpenSARShip	2017	OD	17	Sentinnel-1	11.3K	—	10m	0.87	?	?	—
428	OpenSARShip 2.0	2017	OD	17	Sentinnel-1	34.5K	120	10m	0.87	?	?	—

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
429	PlanesNet	2017	OD	–	OPT (Satellite)	32K	20	–	0.39	?	?	–	
430	RSOD	2017	OD	4	RGB	946	1000	0.3~3m	0.309	?	?	–	
431	SSDD	2017	OD	1	SAR (Satellite)	1.2K	500	1~15m	0.995	?	?	–	
432	WAMI DIRSIG	2017	OD	1	HS (Aerial)	55.2K	64	0.3m	0.327	?	?	–	
433	Airbus Ship Detection Challenge	2018	OD	–	RGB	192.6K	768	–	30	Yes	?	–	Name+Paper
434	DOTA	2018	OD	15	RGB	5.2K	800~4K	–	–	Yes	?	–	TG
435	DOTA v1.0	2018	OD	15	RGB	2.8K	4000	–	1.8	?	?	–	
436	EU Flood	2018	OD	4	OPT (Multiple)	3.7K	1280	–	5.6	?	?	–	
437	HighD	2018	OD	2	RGB	110K	4096x2160	–	–	?	?	–	
438	Iceberg Detection	2018	OD	–	SAR (Satellite)	10K	75	–	0.29	?	?	–	
439	ITCVD	2018	OD	1	RGB	173	5616	0.1m	1.3	?	?	–	
440	Map Challenge	2018	OD	2	RGB	341.1K	300	–	–	Yes	?	–	Name
441	Ships in Google Earth	2018	OD	2	RGB	794	2000	–	2	?	?	–	
442	SpaceNet-4	2018	OD	–	MS+WorldView2	60K	900	0.3m	186	?	?	–	
443	TBF	2018	OD	2	RGB	13	0~40	–	–	Yes	?	–	Paper
444	WHU Building	2018	OD	1	RGB	25.6K	512	0.3m	25	?	?	–	
445	AIR-SARShip-1.0	2019	OD	1	SAR (Satellite)	31	3000	1~3m	0.2	?	?	–	
446	ALCD	2019	OD	–	MS+S2	38	1830	10~60m	0.234	?	?	–	
447	Bridges	2019	OD	1	RGB	500	3822	0.5m	1.45	?	?	–	
448	built-structure-count	2019	OD	–	RGB	5.4K	512	0.3m	2	?	?	–	
449	DLR-ACD	2019	OD	1	OPT (Aerial)	33	4458	–	–	?	?	–	
450	DOTA v1.5	2019	OD	16	RGB	2.8K	4000	–	18	?	?	–	
451	Interaction Dataset	2019	OD	1	RGB	10.9K	–	–	–	?	?	–	
452	Microsoft Building (Australia)	2019	OD	–	–	11.3M	–	–	6.4	?	?	–	
453	MTARSI	2019	OD	1	RGB	9.4K	256	–	0.48	?	?	–	
454	NYCPlanimetric	2019	OD	24	OPT (Aerial)	2.1K	5000	–	2	?	?	–	
455	Oil Palm Plant	2019	OD	2	RGB	20K	–	3m	0.5	?	?	–	
456	Oil Storage Tanks	2019	OD	1	RGB	10K	512	0.5m	3	?	?	–	
457	SAR Ship Detection	2019	OD	1	SAR	43.8K	256	3m-10m	0.407	?	?	–	
458	SAR VD	2019	OD	1	SAR (Aerial)	104	980	–	7.5	?	?	–	
459	SIMD	2019	OD	15	RGB	5K	1024	–	1	?	?	–	
460	Swimming pool/car	2019	OD	2	RGB	3.8K	224	–	0.12	?	?	–	
461	VisDrone2019-DET	2019	OD	10	RGB	10.2K	2000x1500	–	2	Yes	?	–	Paper
462	VisDrone2019-MOT	2019	OD	5	RGB Video	40K	3840x2160	–	14	Yes	?	–	Paper
463	VisDrone2019-SOT	2019	OD	3	RGB Video	139.3K	3840x2160	–	68	Yes	?	–	Paper
464	VisDrone2019-VID	2019	OD	5	RGB	40K	3840x2160	–	14	Yes	?	–	Paper
465	Aerial Maritime Drone	2020	OD	5	RGB	508	800x600	–	0.038	?	?	–	
466	Agriculture-Vision	2020	OD	6	HS (Aerial)	95K	512	0.463m	4.39	?	?	–	
467	AIR-SARShip-2	2020	OD	1	SAR (Satellite)	300	1000	1~3m	0.224	?	?	–	
468	AI-TOD	2020	OD	–	RGB	28K	–	0.3m-30m	22	?	?	–	
469	AU-AIR	2020	OD	8	RGB	32.8K	1920	–	2.2	?	?	–	
470	BIRDSAI	2020	OD	–	TIR	162K	640	–	43.2	?	?	–	
471	DIOR	2020	OD	20	RGB	23.5K	800	0.5m-30m	–	Yes	Yes	–	REO+TG
472	DOTA v2.0	2020	OD	18	RGB	11.3K	4000	0.1~0.81	34.3	?	?	–	
473	DroneRGBT	2020	OD	1	Multiple (Drone)	3.6K	640	–	–	?	?	–	
474	EAGLE	2020	OD	–	RGB	8.8K	936	0.05~0.45m	–	?	?	–	
475	EoC	2020	OD	14	OPT (Aerial)	2.8K	256	–	0.217	?	?	–	
476	FAIR1M	2020	OD	37	OPT (Satellite)	15.3K	5000	–	120	Yes	?	–	TG
477	FGSD	2020	OD	43	OPT (Satellite)	4.7K	930	–	–	?	?	–	
478	HRSID	2020	OD	1	S1+SAR	5.6K	800	0.5~3m	0.581	?	?	–	
479	Intersection Drone	2020	OD	5	RGB	11.5K	4096x2160	–	–	?	?	–	
480	LS-SSDD	2020	OD	1	SAR (Satellite)	15	20000	1~15m	7.8	?	?	–	
481	MOR-UAV	2020	OD	2	RGB	10.9K	608	–	–	Yes	?	–	Paper
482	NEON Tree Crowns	2020	OD	2	RGB	11K	–	–	27.4	?	?	–	
483	Oil and Gas Tank (OGST)	2020	OD	2	RGB	10K	512	0.3m	1.87	?	?	–	
484	RarePlanes	2020	OD	110	MS+WorldView3	713.3K	512	0.3~1.5m	318	?	?	–	
485	Roundabouts Drone	2020	OD	8	RGB	13.7K	4096x2160	–	–	?	?	–	
486	RSOC	2020	OD	4	RGB	3.1K	2500	–	–	Yes	?	–	Paper
487	TGRS-HRRSD	2020	OD	13	RGB	21.8K	930	0.15m-1.2m	9	?	?	–	
488	VisDrone	2020	OD	–	RGB	275.4K	1400	–	85	?	?	–	
489	AFO	2021	OD	2	RGB	3.6K	3840	–	5	?	?	–	
490	Airbus Aircraft Detection	2021	OD	1	RGB	103	2560	0.5m	0.092	?	?	–	
491	Airbus Oil Storage Detection	2021	OD	1	MS	98	2560	1.2m	0.101	?	?	–	
492	AMD	2021	OD	5	OPT (Drone)	508	800	–	0.037	?	?	–	
493	AnimalDrone	2021	OD	–	OPT (Drone)	53.6K	–	–	–	?	?	–	
494	ArcticSeals	2021	OD	1	MS (Aerial)	88K	6576	–	1K	?	?	–	
495	ArtifiVe-Potsdam	2021	OD	1	MS	4.8K	600	0.05m	15.6	Yes	?	–	Paper
496	CASIA-aircraft	2021	OD	2	RGB	58.1K	399	–	–	?	?	–	
497	CASIA-Ship	2021	OD	2	RGB	1.1K	1~680	–	–	?	?	–	
498	CGI Planes	2021	OD	2	RGB	500	–	–	0.7	?	?	–	
499	DroneVehicle	2021	OD	5	RGB-Infrared	56.9K	840	–	–	?	?	–	
500	DSSDD	2021	OD	1	SAR (Satellite)	1.2K	256	–	1.27	?	?	–	
501	Flying Airplanes	2021	OD	1	OPT (Satellite)	8K	–	–	491	?	?	–	
502	Forest Damage	2021	OD	5	RGB	1.5K	1500	–	3.3	Yes	?	–	TG
503	LEVIR-Ship	2021	OD	1	OPT (Satellite)	3.9K	512	–	0.821	?	?	–	
504	MultiDrone	2021	OD	9	OPT (Drone)	259.8K	1280	–	–	?	?	–	
505	NASADEbris	2021	OD	1	OPT (Satellite)	707	256	–	0.073	Yes	?	–	TG
506	NEON Tree Dataset	2021	OD	1	Multiple (Aerial)	2.2K×426	40	–	8.4	Yes	Yes	–	FOMO
507	SaRNet	2021	OD	1	OPT (Satellite)	2.6K	1000	–	0.894	?	?	–	
508	SeabirdsDetection	2021	OD	1	OPT (Aerial)	1	28814	–	2.2	?	?	–	
509	Search and Rescue (SaR)	2021	OD	1	RGB	2.6K	1000	0.5m	–	?	?	–	
510	ShipRSImageNet	2021	OD	1	OPT (Satellite)	3.4K	930	–	4.3	?	?	–	
511	SRSD-v1.0	2021	OD	6	SAR (Satellite)	666	1024	–	0.713	?	?	–	
512	SWIM	2021	OD	1	RGB	14.6K	768	0.5~2.5m	12	?	?	–	
513	TopoBoundary	2021	OD	8	OPT (Aerial)	25.3K	1000	–	20	?	?	–	
514	Urban Tree Crown Detection	2021	OD	1	OPT (Aerial)	220	512	–	0.2	?	?	–	
515	WHUS2-CD+	2021	OD	–	S2	36	10980	10m	27.8	?	?	–	
516	xView3-SAR	2021	OD	2	S1	991×2	27000	10m	2K	?	?	–	
517	Yosemite Tree	2021	OD	1	OPT (Aerial)	1	38400	–	2.7	?	?	–	
518	AIISD	2022	OD	–	RGB	–	–	–	–	?	?	–	
519	BONAI	2022	OD	1	RGB	3.3K	1~24	0.3m-0.6m	4.86	?	?	–	

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
520	Buildings Containers Cranes	2022	OD	3	OPT (Satellite)	—	2500	—	3	?	?	?	—
521	DroneCrowd	2022	OD	1	OPT (Drone)	33.6K	1920	—	10.4	?	?	?	—
522	HRPlanesv2	2022	OD	1	OPT (Satellite)	2.1K	4800	—	8.1	?	?	?	—
523	MSAR	2022	OD	4	SAR (Satellite)	28.4K	256	—	2.5	?	?	?	—
524	OnlyPlanes	2022	OD	1	OPT (Satellite)	12.5K	1024	—	15	?	?	?	—
525	RampBuilding	2022	OD	1	OPT (Satellite)	96.3K	256	—	24.8	?	?	?	—
526	RSDD-SAR	2022	OD	—	SAR (Satellite)	7K	512	—	—	?	?	?	—
527	SADD	2022	OD	1	SAR (Satellite)	3K	224	—	—	?	?	?	—
528	SAR Aircraft	2022	OD	2	SAR	3K	224	0.5m-3m	0.18	?	?	?	—
529	Satellite Pool Detection	2022	OD	1	OPT (Satellite)	14.9K	512	—	3.7	?	?	?	—
530	SeaDronesSee	2022	OD	6	RGB	5.6K	3~840	—	60.3	Yes	?	?	Paper
531	SODA	2022	OD	9	RGB	2.5K	2.7K~4.8K	—	—	Yes	?	?	TG
532	UAVOD-10	2022	OD	10	OPT	844	—	—	—	Yes	?	?	Paper
533	VISO	2022	OD	4	OPT (Satellite)	15.5K	12000	—	8.6	?	?	?	—
534	MDMT	2023	OD	—	OPT (Drone)	39.7K	—	—	25.8	?	?	?	—
535	EverWatch	2024	OD	8	RGB	5.3K	1500	0.01m	—	Yes	Yes	Yes	GB2+TG
536	LEVIRDet-159	2026	OD	159	OPT	700K	—	—	—	?	?	?	—
537	MSDS900	—	OD	1	OPT (Satellite)	3K	900	—	—	?	?	?	—
538	MSDSfull	—	OD	1	OPT (Satellite)	200	4800	—	—	?	?	?	—
539	PSDS	—	OD	1	OPT (Satellite)	1.3K	900	—	—	?	?	?	—
540	SMCDD	—	OD	—	SAR (Satellite)	—	—	—	—	?	?	?	—
541	Sydney caption	2016	IC	7	RGB	613	500	0.5m	0.298	?	?	?	—
542	UCM caption	2016	IC	21	RGB	2.1K	256	0.3m	0.255	?	?	?	—
543	RSICD	2017	IC	—	OPT+Text	10.9K	—	—	—	?	?	?	—
544	RSITMD	2021	IC	—	OPT+Text	4.7K	—	—	—	?	?	?	—
545	LEVIR-CC	2022	IC	—	RGB	10.1K	512	—	—	?	?	?	—
546	ChatEarthNet	2024	IC	—	S2+Text	173.5K	256	10m	38.4	?	?	?	—
547	LHRS-Align	2024	IC	—	RGB	—	—	0.5m-30m	—	?	?	?	—
548	VRSBench	2024	IC	—	OPT+Text	29.6K	—	—	—	Yes	Yes	Yes	REO
549	RSCC	2025	IC	—	OPT+Text	62.4K	—	—	—	?	?	?	—
550	BigEarthNet.txt	2026	IC	—	S1/S2+Text	464K	—	10m	—	?	?	?	—
551	Sentinel2Cap	2026	IC	—	S2+Text	—	—	—	—	?	?	?	—
552	DFC06	2006	SR	—	MS (Aerial)	6	5000	—	0.39	?	?	?	—
553	Proba-V Super Resolution	2018	SR	—	MS (Satellite)	1.2K	384	—	0.692	?	?	?	—
554	SEN12MS-CR	2020	SR	—	S1/S2	122.2K	256	10m	—	Yes	Yes	Yes	GFM+Paper
555	ALSAT-2B	2021	SR	—	OPT (Satellite)	5.5K	256	—	0.07	?	?	?	—
556	PAirMAX (Airbus)	2021	SR	—	MS (Satellite)	14	—	—	0.153	?	?	?	—
557	PAirMAX (MAXAR)	2021	SR	—	OPT	14	—	—	—	?	?	?	—
558	PushFrameSR	2021	SR	—	OPT (Satellite)	960×15	256	—	1.1	?	?	?	—
559	Jilin1 VidSR	2022	SR	—	OPT (Satellite)	6.2K	3000	—	19.4	?	?	?	—
560	MultiExposureSR	2022	SR	—	OPT (Satellite)	—	—	—	0.743	?	?	?	—
561	Mu-S2	2022	SR	—	MS (Satellite)	1.3K	100	—	2.8	?	?	?	—
562	PRISMA Pan Sharp	2022	SR	—	HS (Satellite)	4	2400	—	1.7	?	?	?	—
563	SEN2VENμS	2022	SR	—	S2+VENμS	133K	256	10m-20m	85	?	?	?	—
564	WorldStrat	2022	SR	8	Multi	3.4K	1054	1.5m-60m	110	Yes	?	?	TG
565	Africa MiniCubes	2021	P	—	Multi	—	—	—	—	No	No	No	—
566	SeCo	2021	P	—	S2	200K×5	264	10m	43.6	No	No	No	—
567	Hephaestus	2022	P	—	SAR	110.6K	—	—	—	No	No	No	—
568	SSL4EO-S12	2022	P	—	S1/S2	251.1K	—	—	—	No	No	No	—
569	Clay v1.5 pretraining corpus	2024	P	—	Multi	70M	156x256	0.5m-500m	—	No	No	No	—
570	DOFA-data (Metadataset)	2024	P	—	Multi	8.1M	—	0.5m-30m	12K	No	No	No	—
571	EarthDial	2024	P	—	Multi	11.1M	—	—	—	No	No	No	—
572	GRAFT	2024	P	—	RGB+MS	—	—	—	—	No	No	No	—
573	MajorTOM-Core	2024	P	—	S1/S2+DEM	1.5M~2.2M	356~1.1K	10m~30m	63K	No	No	No	—
574	MMEarth	2024	P	—	Multi	1.2M	128	10m	597	No	No	No	—
575	Prithvi-EO-2.0 pretraining corpus	2024	P	—	HLS (L30/S30)	4.2M	256	30m	—	No	No	No	—
576	Satelloptic (EarthView)	2024	P	—	Multi	—	—	1m-40m	—	No	No	No	—
577	SkyScript	2024	P	—	RGB+Text	5.2M	—	0.1m-30m	—	No	No	No	—
578	SkySense	2024	P	—	Multi	—	—	0.31m-20m	—	No	No	No	—
579	SpectralEarth	2024	P	—	EnMAP HS	539K	128	30m	3.3K	No	No	No	—
580	Copernicus-Pretrain	2025	P	—	Multi	18.7M	—	—	—	No	No	No	—
581	GeoPlex	2025	P	—	Multi	5 datasets	—	0.2m-250m	—	No	No	No	—
582	SAR-GEOVL-1M	2025	P	—	SAR	120K	—	—	—	No	No	No	—
583	SkySense++ pretraining corpus	2025	P	—	Multi	27M	—	0.5m-30m	—	No	No	No	—
584	SSL4EO-S12 v1.1	2025	P	—	S1/S2	246.1K×4	—	10m	—	No	No	No	—
585	TerraMesh	2025	P	—	Multi	9M	264	10m-30m	31K	No	No	No	—
586	SpectralEarth-MM	2026	P	—	Multi	2M loc.	—	—	40K	No	No	No	—
587	GEO-Bench	2023	B	—	Multi	12 datasets	—	—	~65	Yes	Yes	Yes	GB1+Name
588	FoMo-Bench	2024	B	—	Multi	15 datasets	—	—	—	Yes	Yes	Yes	FOMO+Name
589	PANGAEA	2024	B	—	Multi	11 datasets	—	—	—	Yes	Yes	Yes	Name+PG
590	Copernicus-Bench	2025	B	—	Multi	15 datasets	—	—	—	Yes	Yes	Yes	CB+Name
591	PlantationBench	2025	B	—	Multi	—	—	—	—	Yes	Yes	Yes	Name+PB
592	EarthShift	2026	B	—	Multi	11 tasks	—	—	—	Yes	Yes	Yes	ES+Name
593	MMEarth-Bench	2026	B	—	Multi	5 tasks	—	—	—	Yes	Yes	Yes	MMB+Name

Flags: Benchmark / FM Eval: Yes = confirmed; No = confirmed negative; ? = not yet established. Evidence: TG = TorchGeo; GB1 = GEO-Bench; GB2 = GEO-Bench-2; GFM = documented GeoFM/FM evaluation; PG = PANGAEA; FOMO = FoMo-Bench; CB = Copernicus-Bench; REO = REOBench; LB = Landsat-Bench; PH = PhilEO Bench; PB = PlantationBench; ES = EarthShift; MMB = MMEarth-Bench; Paper = scientific paper; Name = explicit benchmark/challenge/evaluation naming.